Enhanced tracking and reporting of missing persons using semantic web technologies

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***Abstract*—The issue of missing persons is a critical concern in the United States, where thousands of individuals are reported missing each year. There is a need to find more effective ways of integrating and analyzing the data in order to recognize the patterns of disappearances. This research attempts to leverage the robustness of semantic web models and knowledge graphs to provide a comprehensive and interconnected view of missing person data. Ontologies are a powerful data representation tool to connect real world entities into a unified model. These tools can provide a unique way of drawing connections as to why Americans go missing. Knowledge graphs will be constructed around details pertaining to victims, which include the circumstances of disappearance, characteristics of the victims, and locations where the victims were last seen. The knowledge graphs constructed will also serve as a foundation for software tools that support real-time analysis, aiding in proactive intervention and case resolution.**

***Index Terms*—Semantic web and Semantic web models, Ontologies, Knowledge Graphs, Missing people, Risk factors, NamUS.**

I. Introduction

To begin, we will need to first define who a missing person can be, the causes for people to go missing, and how we can use a semantic web model to form knowledge graphs that can be used to bridge connections with data in regard to missing people. By defining ontologies around the concept of missing persons, potential causes, and related contextual data, the project aims to create a unified knowledge graph that bridges various data sources. This will allow us to define use cases for the software we will construct over the course of our research.

A. Defining Missing People and Potential Causes For People To Go Missing

Howlings et al. [1], describes a missing person as an individual who cannot be located nor can have their well-being confirmed. There are diverse causes of people going missing highlights that it is unlikely that there is a single driving cause for it to occur, but there are multiple risk pathways that can put a victim at risk [9]. We can see a need to develop a system that combines or links these risk factors to an existing database of missing persons to help facilitate their search and rescue.

B. Semantic Web, Ontologies, and Knowledge graphs

This is where semantic web models come into play. A semantic web model builds an ontology. These ontologies represent the semantics that are attached with domain concepts, which in turn uses ontological terms to define real-world objects [4]. Ontologies are structured models that create a shared understanding of a specific domain, helping people and software communicate consistently. They define groups of entities (or classes) and their relationships, creating a clear hierarchy where subclasses inherit the properties of their superclasses. Ontologies also let developers define specific relationships, such as whole-part or causality, to explain how entities relate and interact.

Ontologies are valuable for sharing information, aligning perspectives across teams, supporting tool integration, and enabling automated reasoning. They provide a common language for different systems, ensuring data consistency and allowing smooth data validation. In complex, interdisciplinary fields, like safety analysis, ontologies help standardize information and make cross-team collaboration easier [3].

Ontologies utilize knowledge graphs, and explicit formal conceptual models, which can be put together to form connections and discover new data, which will be explained further down in the problem definition.

II. Problem definition

According to the World Population Review, the United States has the highest number of reported missing persons cases in the world. Given its robust reporting system and large population, we can construct an ontology to track these reports, taking into account factors such as age, gender, and race of the missing individuals along with their reason of going missing.

III. Related Literature

As the research delves into uncovering patterns as to why victims may go missing, potential categories need to be identified first in order to provide direction in noticing the patterns. Howlings et al. [1], in their research on the use of social media by the police for investigating missing children, have listed a variety of reasons that can be attributed to why people go missing, ranging from escaping abusive living conditions, escaping from stressful environments, or physical and mental health issues [1].

Henderson et al. [2], an Australian-based research team, conducted surveys to further list other potential reasons why people may go missing. The survey results categorize the

TABLE I
USE CASES TABLE

Use Case	Description (This use case refers to retrieving of data of missing people-)	Input	Output
Age-based results	based on age	Age range	All data related to people fitting the specified criteria: Case Number, DLC (last contact date), Personal details (e.g., Name, Missing Age), Location (City, County, State), and attributes (e.g., Sex, Race/Ethnicity).
Location-based results	based on last seen location	Location/ Area/ State	
Race-based results	based on race	Race	
City-based results	based on city of origin	City	
County-based results	based on county of origin	County	

causes for people to go missing into: unintentionally wandering off or being lost, attempting to run away from consequences, rebelling against an authoritative figure (most likely parents), and safety concerns, which can relate to kidnapping and harm [2]. Given these categories, an attempt can be made to connect them to the victims in order to glean new information from our knowledge graphs.

For the scope of the project, we will gather data from NamUs. NamUs is a government-run resource center for missing, unidentified, and unclaimed persons within the United States [11]. The website provides detailed information on missing people reported per state. For our research, we considered three different states, California, Texas, and Alaska. The World Population Report keeps a tracker on the number of people that go missing in the US. The report states that California, Texas, and Florida are the three states with the most open cases of missing people to be recorded [7]. Naturally, this would mean including Florida as the third state, but, an interesting metric from the World Population Report made us consider Alaska, as Alaska has the most number of people going missing per 100,000 people [7].

With further deliberation, the research team decided to focus on the state with the highest number of missing people reported annually, which is California. According to the World Population Review, California leads the US in the number of missing people, with 3,010 individuals reported missing in 2024 alone. This will give us a rich set of data to construct ontologies as we can take each person as a real-world object to create knowledge graphs on. Yahya et al. [4] demonstrate the effectiveness of using Knowledge Graphs in real-world contexts. Their research on using Knowledge Graphs for Industry 4.0 (I4.0) applications delved into building a reference generalized ontological model (RGOM) which they have shown can ”can facilitate a range of I4.0 concepts including improved asset monitoring, production enhancement, reconfiguration of resources, process optimizations, product orders and deliveries, and the life cycle of products.” [4], which proves the effectiveness of knowledge graphs in capturing the minutia of real-life objects.

Furthermore, on our choice of California, a study by Johnson et al. [6] explains that California as a state possesses a diverse population. According to the survey and research conducted by the Public Policy Institute of California, no race or ethnic groups constituted a majority of California's population [6]. This can prove advantageous for us, as the diversity provides us with an even richer set of data to draw from given the distribution which can be tied to our use case of categorizing based on ethnicity.

By constructing this framework, we can uncover potential reasons why people within a certain state go missing. Gomes Jr., et al. [5]. constructed a framework to gather information on missing persons in Brazil and discovered age as a factor for why Brazilian citizens go missing, citing that ”The disappears (sic) of a person according to age may be related to several factors such as slave labor, chemical dependency, and involvement with debts loss of memory”. This shows that there is a precedence for building a framework that can procure new knowledge. This is consistent with the findings from Howlings et al. [1]. who have listed physical and mental health issues as being the cause of victims going missing [1]. This also matches with the findings of Henderson et al. [2], who have categorized escaping adverse conditions such as financial difficulties under its own category. Szekely et al. [10]. developed DIG, a knowledge graph system designed to combat human trafficking by using semantic technologies to formulate connections with victims of human trafficking. Missing persons reports can be linked to safety concerns as highlighted by Henderson et al. [2]. further validating the approach of previous research teams in building semantic web models to derive new insights from missing person reports.

IV. APPROACH AND HIGH-LEVEL SYSTEM DESIGN

Our approach begins with retrieving a collection of information on all current missing persons cases in the state of California. The NamUS website provides a downloadable CSV file of all such cases which contains information on attributes related to the victim, such as the victim's case's number, date of last contact (DLC), name, missing age, City, County, State, Biological sex, and Race/Ethnicity. The website also allows its user to categorize by state, drastically reducing the scraping needed to keep it secluded to certain states.

With the CSV data for California retrieved, the next step was to scrape for data that is dynamically generated within the website due to APIs. The website makes API calls that display specific information attached to a victim. This information is not statically present in the NamUS website nor is it provided in the downloadable CSV, which meant that the usual tools used to scrape data such as BeautifulSoup would not work. Examples of information of this kind include the circumstances of the disappearance of the victim, current age, location of last contact and their physical description. The API is not publicly accessible either, as the more minute details

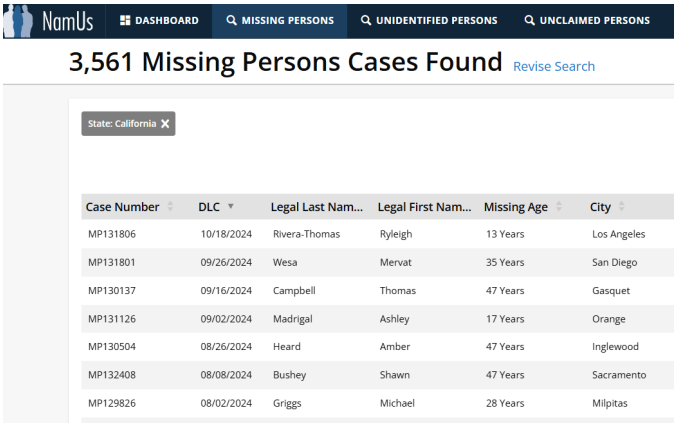


Fig. 1. The NamUS dashboard, retrieving California data.

pertaining to missing victims fall mostly under government domain. This information would be important to include, as it could be used to classify victims based on common themes/circumstances.

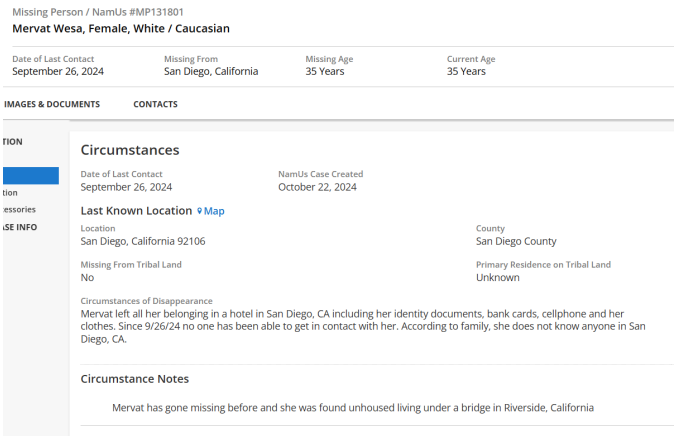
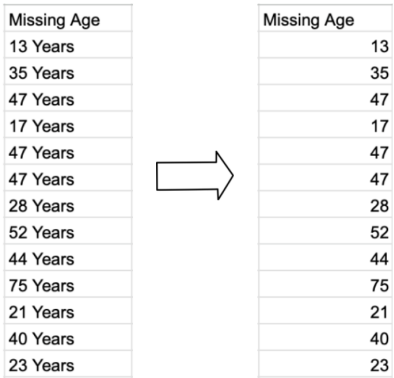


Fig. 2. Typical victim profile that's available from the website which contains information not immediately available in the downloadable CSV file, such as circumstances of disappearance

Therefore, Selenium, a browser automation tool, was used to load individual profiles and get the information generated dynamically by the API in order to collect and analyze this data. To help pinpoint the last known locations of missing persons, the reported location was included as a separate feature, with a link to Google Maps for quick access to their last point of contact. This link was extracted and added to the CSV file, along with a dedicated column for latitude and longitude coordinates. This setup not only makes it easier to visualize each case but also aids in clustering similar disappearance cases based on location. This process was repeated for Alaska and Texas as well in the event we wished to use their data for comparisons with California.

Following the retrieval of data from missing persons in California, the next step was to clean the data to make it more relevant to our use cases. The first issue with the retrieved dataset was that the ages were presented in full phrases i.e.



Missing Age
13 Years
35 Years
47 Years
17 Years
47 Years
47 Years
28 Years
52 Years
44 Years
75 Years
21 Years
40 Years
23 Years

Missing Age
13
35
47
17
47
47
28
52
44
75
21
40
23

Fig. 3. Missing Ages processed, before and after

"13 Years", and "21 Years" as opposed to being presented as an integer. This was done with both Missing Ages (the ages when victims went missing) and Current Ages (the ages which represent the current age that a victim would be if they were found today). Python was used to convert the ages as necessary into integers so that data retrieval based on age categories could be done more easily.

If a victim went missing and if they were less than a year old, their ages were displayed as '$<$ 1 Year' in the CSV. Getting the exact date of birth and using that to retrieve actual ages proved to be difficult for the team, so we made the decision to round their ages down to 0. This would still allow them to be within their own category of victims less than a year old.

The next step of pre-processing the data was to combine the names. The names were in separate columns in the scraped data, consisting of 'Legal Last Name' and 'Legal First Name'. In the final output, it would be easier to keep the names merged as one to prevent mismatching if we wished to sort the data alphabetically. Furthermore, it wouldn't be relevant to try to pattern match based on names.

Next, we wanted to include the URLs for each individual case as a potential URI (Uniform Resource Identifier). The case numbers map to the individual URLs. The URLs are structured as follows: `https://www.namus.gov/MissingPersons/Case#/[`*`case number for the victim`*`]?nav`. In the scraped data, the case numbers are included as MP [*case number for the victim*]. In the pre-processing, it was a matter of combining the strings in the right manner to map the appropriate URL which was then included as its own column.

The scraping also retrieved information about the victim's physical description which included distinctive characteristics like tattoos and scars. This information would not be useful in uncovering patterns, and would likely introduce complexity to our knowledge graphs, so it was removed in the cleaned data.

The scraped data also included date modified, which would give the user the information on the last update a particular case may have received. At its present iteration, it would

TABLE II
ATTRIBUTES FROM THE CLEANED TABLE THAT WILL BE USED IN THE ONTOLOGY

Relevant Attribute	**Description of Attribute**
Case Number	The unique identifier attached to a case
DLC (Date of Last Contact)	The last time the victim was contacted before they went missing
Legal Name	The legal name of the victim
Missing Age	The age that the victim went missing
City	City of origin for the victim
County	County of Origin for the victim
Biological Sex	The Biological Sex of the victim
Race / Ethnicity	The race and ethnicity of the victim
Current Age	The current age of the victim at the present data
Map Data	Google Map location of where the victim was last seen.
Latitude, Longitude	Coordinates of the place the victims were last seen
Circumstances of Disappearance	The last reported circumstances of the victim before they went missing.
URL	The hyperlink to the victim's NamUS profile and related case

be challenging for our platform to dynamically update the knowledge graph for individual cases, so this was not included in the cleaned data.

The last step in our pre-processing included the removal of the state column. Since we were handling particular American states, we felt that the inclusion of the state column wouldn't be necessary as it would only repeat the information that was already established.

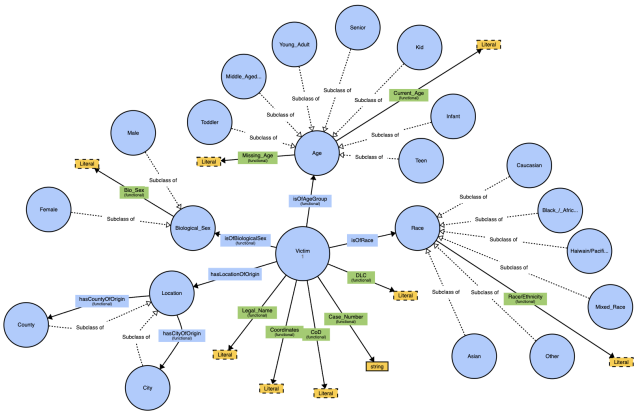


Fig. 4. Visualized ontology using WebVOWL

The missing persons ontology is a detailed system used to organize information about missing persons. It includes different categories like race (Asian, Black / African American, Caucasian) and important details such as case numbers registered with NamUS, locations from where the person went missing and the circumstances leading to their disappearance missing. Each case has unique properties like the case number, city, county, and coordinates of the location. It also keeps track of specific information like the last time the person was contacted, their legal name, their age now, and their age when they went missing. This system can help law enforcement and search teams find and manage information quickly and accurately.

The user interface that we will implement will be a Web App that allows the user to query based on the categories of our knowledge graph. If, for instance, a user wants to analyze a trend of victims with regards to a certain race, our website will provide functionality that makes a query based on the selected race.

To make sure we have a clear implementation plan of tasks that need to be completed we have decided to follow Agile/Scrum Project Management Methodology, with the help of the open source tool Taiga Board.

For our overall implementation plan, we decided on the following:

- Fetch and refine data for pre-processing
- Build knowledge graph based processed data
- Write SparQL queries to fetch required data after constructing the knowledge graph
- Build the website for users to make the queries and see the results, whilst uncovering patterns to missing people

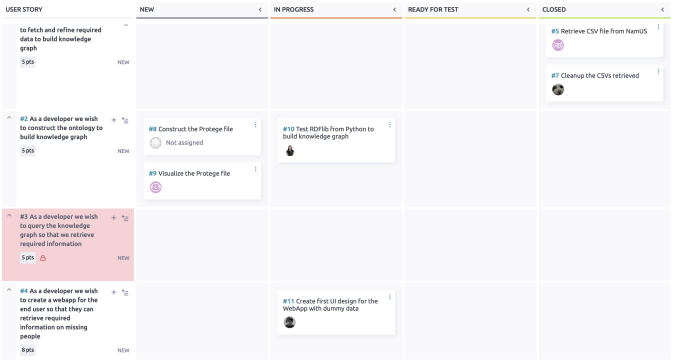


Fig. 5. Scrum Board on Taiga

We created User Stories that help us break down the process into more manageable chunks. This helps us delegate the tasks at a more granular level and track their progress in case there are any blockers or spike activity that needs to be conducted. This will in turn help us complete the tasks on time, ensuring work is done consistently throughout the project's life-cycle, and allows to account for new changes.

- Suparno Roy Chowdhury and Mustakim Shikalgar worked on preprocessing the data.
- The whole team was responsible for creating the Ontology.
- Ramchander Venugopal created the visualization of the Ontology.
- Mrunal Kapure worked on the initial creation of the Knowledge Graph.
- Ayush Desai created the Initial Web Application Setup and will be working on building the initial User Interface.

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